SYNOPSIS

Title

Real-Time Sports Analytics Feed & Winner Predictor

Introduction

A web-based application that analyzes historical and live sports data to provide real-time analytics and predict the winning team using Machine Learning. The project helps users understand team and player performance through interactive dashboards.

Problem Statement

Existing sports platforms mainly provide live scores and statistics but do not offer intelligent winner prediction and comprehensive analytics in one place.

Objectives

- Develop a real-time sports analytics dashboard.
- Predict match winners using Machine Learning.
- Display team and player statistics.
- Visualize historical and live match data.
- Provide an easy-to-use web interface.

Scope

- Covers sports analytics, team comparison, and winner prediction.
- Intended for sports fans, analysts, students, and coaches.
- Initial implementation supports one sport.

Proposed Solution / Methodology

- Collect historical sports datasets.
- Preprocess and clean the data.
- Train Machine Learning models (Logistic Regression/Random Forest).
- Build REST APIs for prediction.
- Display analytics and predictions through a web dashboard.

Features / Modules

- User Authentication
- Dashboard
- Team Analytics
- Player Analytics
- Winner Prediction
- Historical Statistics
- Admin Panel

Technology Stack

Programming Languages: Python, JavaScript
Frameworks: React.js, Node.js, Express.js
Database: PostgreSQL
Libraries: Scikit-learn, Pandas, NumPy, Chart.js
Tools: VS Code, Git, Postman

System Requirements

Hardware: Intel Core i5, 8 GB RAM, 256 GB SSD
Software: Windows 10/11, Python 3.11+, Node.js, PostgreSQL
Browser: Chrome, Edge, Firefox

Expected Outcomes

- Predict match winners with good accuracy.
- Provide real-time sports analytics.
- Display interactive dashboards and player statistics.

Future Scope

- Multi-sport support.
- Mobile application.
- AI-generated match summaries.
- Fantasy team recommendations.

Conclusion

The project combines Machine Learning and full-stack web development to provide real-time sports analytics and intelligent winner prediction through a user-friendly web application.